



Fig. 3: Wire library on the left panel in Logisim (Courtesy: softspeda)

and a finger, and change values in the circuit. So, if you click an input value of 0 with the finger, it will change to 1, meaning the input is receiving electric flow.

Cursor is the default mouse pointer. You can use it for interactive actions on the workspace, like selecting and moving components, connecting or deleting wires, resetting component properties and more. The icon for textbox lets you add labels to your circuit for convenience.

Following the icons are symbols of five of the most used components in an electronic circuit—input, output, as well as NOT, AND OR logic gates. Use the cursor to drag these elements on the toolbar and drop them on the workspace for use in the circuit as desired.

The left panel of the software consists of libraries and properties windows. The first folder represents the project. It is followed by folders called Base, Gates, Plexers, Arithmetic, Memory and Input/Output. All folders contain components from the library that can be used for building different kinds of circuits and sub-circuits. These components are categorised in folders for easy retrieval.

For instance, under Gates folder, along with the common three logic gates mentioned earlier, you get NAND, NOR, XOR, XNOR, Buffer, Odd Party, Even Party and so on.

Below the folders comes Properties window. Here you can edit

properties of all components. The window displays circuit name, text label assigned, direction and font of the label, number of input ports of logic gates and so on. To modify any component, just click the component in the circuit with the cursor and do the modifications in this window. Suppose you dragged in an AND gate that has two input ports by default. If you need four input ports, click the AND gate symbol and in the properties window, click to change the number in Inputs entry field. The logic gate in the circuit will update immediately.

The best part

Logisim not only lets you simulate your logical circuits but also edit circuits while simulation is running. Therefore you can immediately rectify flaws without the need of separate edits and loading. Another useful feature is that any circuit built in Logisim software can be used as an existing library component. That means, you can use the created circuit as a sub-circuit of another bigger one. In addition, you can save an image file of the circuit created. Go to File→Export Image and then select the file type of your choice.

There are many more features that you can explore in Logisim—be it the different components or the different simulation settings. Go to Tutorial under Help tab to find the desired information. **EFY**

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